A380X VD airports as flat ground

The A380X VD terrain profile shows the airports around the aircraft as flat ground at their elevation instead of spikes, and the ND / VD map views survive flight reloads

Aircraft	A380X (the map view fix: A32NX and A380X)
Where to find it	VD under the ND (terrain profile); ND weather and terrain
Branch	`core/a380x-vd-runways`, based on `core/nd-weather-terrain` with `core/oans-btv-raas` merged in (stacked: needs both merged first; the airports come from the nearby runway search of `core/oans-btv-raas`, merged again for its airports-with-a-runway fix).
Status	Airport circles: tested in the sim by the user: OK (one circle at RJTT); the empty list after take-off seen there is fixed and passed (group 2). The first version (runway rectangles) was tested. Map views across flight reloads: VD OK after 4 flight reloads, but the leak did not occur in that test, so the fix itself is not proven in the sim.
Pictures	No pictures.

1. VD airports drawn as flat ground

> **Aircraft:** A380X | **Where:** VD terrain profile | **Status:** Tested in the sim by the user: OK (one circle of radius 1.8 NM at RJTT, flat ground, no spikes). The airport list emptied at about 2,900 ft after take-off in that test; fixed by the runway search of `core/oans-btv-raas`, then OK (group 2 : 10 airports on the ground, 11 to 12 in the climb). The earlier runway rectangles: tested (VHHH OK, one short spike at CYUL). Unit tests: VdAirports.spec, ndwxr vd_airports_test.

What it is for

The MSFS terrain map draws the airport diagram in light grey over the terrain, and no setting hides it. The VD read that grey as terrain far above its plot: spikes wherever the vertical cut crossed an airport, the destination on approach included.

- The systems host searches the sim airports within 25 NM that have a land runway, and publishes up to 24 airports, closest first, as one circle each: the smallest circle around the corners of all its runways, plus 0.3 NM for the taxiways and aprons beside them, at the elevation of its highest runway threshold.
- Where the cut crosses such a circle, the VD draws flat ground at the airport elevation instead of the terrain map.
- The first version used a rectangle around each runway (plus 0.15 NM): taxiways and aprons between and beside the runways still showed as spikes. One circle per airport covers them (the user chose it over a fixed radius).

What you should see

Situation	What you should see
VD cut across an airport	Flat brown ground at the airport elevation over the whole airport (runways, taxiways, aprons), no spike.
Past the edge of the airport circle	The real terrain again.
Approach to the destination	The airport ahead shows as flat ground at its elevation.
After take-off, climbing away	The departure airport stays in the list (and flat) while it is within 25 NM.
Flight path and MSA line over an airport	The green path, the pink MSA / MORA line and the symbols stay visible over the flat ground (they used to be hidden over the first NM at the departure airport).

Limitations and why

Limitation	Why	What to do
Real terrain inside an airport circle is flattened (for example a hill beside the runways).	Design choice: an airport is drawn as flat ground; the circle is the smallest one around the runways plus 0.3 NM.	-
Airports beyond 25 NM, or more than 24 airports, are not flattened.	Design choice of the search radius (enough for the VD ranges of the approach) and of the gauge list size.	-
Airports without a land runway of 100 m or more are not flattened.	The runway search leaves out heliports and water-only bases (`core/oans-btv-raas`), so they no longer push the real airports out of the list.	-

How to use

1. Nothing to set: read the VD terrain profile as usual (TERR SYS of the SURV page not OFF).

How to test

#	Do this	Expected result
1	A380X at a gate of an airport with parallel runways (for example CYUL or RJTT), flight plan from there. Both NDs ARC, range 10.	On the VD, flat brown ground at the airport elevation over the whole airport (no grey spikes from taxiways or aprons); past the airport edge, real terrain again.
2	Taxi, line up and take off; climb to 3,000 ft and beyond.	Flat ground under and ahead of the aircraft along the runway; after take-off the departure airport stays flat on the VD (the list does not empty).
3	A380X on approach to any airport, the VD cut crossing it.	The airport shows as flat ground at its elevation; no tall spikes.

2. ND and VD map views across flight reloads

> **Aircraft:** A32NX and A380X | **Where:** ND weather radar / terrain and A380X VD (native ndwxr gauges) | **Status:** VD OK after 4 flight reloads (user). In that test the sim restarted the module at each reload, so the leak this fixes did not occur: the fix itself is not proven in the sim. Unit test: ndwxr gauge_slots_test.

What it is for

The ND weather and terrain and the A380X VD are drawn by native gauges (ndwxr) on sim map views. Over flight reloads the gauges could leak their map views: the module keeps its state while the next flight's gauges start before the old ones are closed. Once the sim's view budget was used up, the VD went blank.

A slot table now hands each gauge over: a new gauge of the same display or role takes over or replaces the previous flight's gauge, deletes its views (never a view another gauge of the same display still uses, as view numbers are per display), and keeps the module within its view budget.

Limitations and why

Limitation	Why	What to do
The original blank VD after a reload was not reproduced.	In the reload test the sim restarted the module each time, so no old view was left; the blank VD seen before may come from another reload path.	Report the reload path (flight reload, return to the main menu, new flight) if the VD goes blank again.

A full MSFS restart is needed after an update.	The native module (ndwxr.wasm) changes.	Restart the sim after installing.

How to use

1. Nothing to set: reload the flight as usual; the ND weather / terrain and the VD come back.

How to test

#	Do this	Expected result
1	A380X at a gate, aircraft powered, both NDs ARC, TERR ON ND on one side, the VD shown. Note the VD and ND pictures.	Terrain on the ND and the VD profile.
2	Reload the flight (the same airport) four times, checking the NDs and the VD after each reload.	After every reload the ND terrain / weather and the VD profile come back; the VD never stays blank.
3	Repeat once on the A32NX with TERR ON ND.	The ND terrain comes back after each reload.

3. VD profile starts at the aircraft on the ground

> **Aircraft:** A380X | **Where:** VD under the ND (green flight path) | **Status:** Tested in the sim by the user: the path now matches the F-PLN from the ground (RJAA, 1,480 ft in the F-PLN at 1 NM). Unit test: VdProfileStart.spec. The 1,620 ft the VD reads at 1 NM against 1,480 ft in the F-PLN was not explained.

What it is for

The FMS profile starts at the take-off point of the flight plan. An aircraft parked or taxiing past it has that point behind it, so the first part of the climb fell off the left edge of the VD: the green path seemed to start at about 1,500 ft.

On the ground the whole profile is now moved back so that the take-off point sits at the aircraft (0 NM): the path rises from the ground and keeps the distances the FMS predicts.

What you should see

Situation	What you should see
On the ground, aircraft past the take-off point of the plan	The green path rises from the ground at 0 NM and matches the F-PLN altitudes and distances.
In the air, or the take-off point ahead of the aircraft	No change.

Limitations and why

Limitation	Why	What to do
On the ground the path is shifted by the distance to the take-off point; waypoint symbols and the grey area keep their real distances.	Design choice: only the profile is moved, so the climb is complete at the aircraft.	-

How to use

1. Nothing to set: enter the flight plan with departure, SID and transition and read the VD on the ground.

How to test

#	Do this	Expected result
1	A380X on the ground, flight plan with departure, SID and transition, ND ARC, VD range 10.	The green path rises from the ground at 0 NM to about 1,500 ft at 1 NM; it reaches 10,000 ft near 12 NM (range 20).